International University

School of Computer Science and Engineering

Artificial Intelligence

Lab 4 & 5: Sudoku (CSP)

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1. Introduction

1.1 Performance Scoring

Sudoku is a classic constraint satisfaction problem (CSP) in which a 9×9 grid must be filled with digits 1–9 such that each row, column, and 3×3 sub-box contains all digits exactly once. A standard puzzle provides a partial assignment (the "clues"), and the goal is to find a complete, consistent assignment.

This lab implements an AI agent that solves Sudoku puzzles by formulating the problem as a CSP and applying two complementary algorithms:

- AC-3 (Arc Consistency 3) — a constraint propagation technique that prunes infeasible values from variable domains.
- Backtracking Search with Forward Checking — a depth-first search enhanced with heuristics and inference to explore the solution space efficiently.

1.2 Objectives

1. Model Sudoku as a formal CSP with variables, domains, and constraints.
2. Implement the AC-3 algorithm for domain reduction.
3. Implement Backtracking Search with MRV heuristic, Forward Checking, and domain restoration.
4. Evaluate the solver on two benchmark datasets: Project Euler (50 puzzles) and Magic Tour (95 puzzles).
5. Build an interactive GUI for step-by-step visualization of the solving process.

2. CSP Formulation

2.1 Variables

The Sudoku grid is represented by 81 variables, one per cell. Each variable is identified by a two-character string combining a row letter (A–I) and a column digit (1–9):

```
Variables = {A1, A2, ..., A9, B1, B2, ..., I9}   (|Variables| = 81)
```

The variables are generated using the cross product utility:

```
rows = "ABCDEFGHI"
cols = digits = "123456789"
squares = cross(rows, cols)  # ['A1', 'A2', ..., 'I9']
```

2.2 Domain

Each variable's domain is the set of digits it can take. For clue cells (given values), the domain is a singleton; for empty cells, the initial domain is {1, 2, 3, 4, 5, 6, 7, 8, 9}:

```
def getDict(self, grid=""):
    for cell in self.variables:
```

```
        if grid[i] != '0':
            values[cell] = grid[i]      # singleton domain
        else:
            values[cell] = digits        # full domain "123456789"
```

Domains are stored internally as strings for efficient manipulation (e.g., "123456789", "37", "5").

2.3 Constraint

The constraints enforce the all-different property across three types of units:

Unit Type	Count	Description
Row	9	All 9 cells in a row must be different
Column	9	All 9 cells in a column must be different
Box	9	All 9 cells in a 3×3 sub-box must be different
Total	27	

For each variable, its peers are all other variables that share at least one unit with it. Each cell has exactly 20 peers (8 from its row + 8 from its column + 4 additional from its box).

The binary constraints are explicitly enumerated as pairs:

```
self.constraints = {(variable, peer)
                    for variable in squares
                    for peer in self.peers[variable]}
```

This gives 1620 binary constraint arcs (81 variables × 20 peers each).

3. Algorithm Design

3.1 AC-3 Algorithm

AC-3 enforces arc consistency by iteratively removing values from variable domains that cannot participate in any consistent assignment with a neighboring variable.

Implementation

```
def AC3(csp):
    queue = list(csp.constraints)
    while queue:
        (xi, xj) = queue.pop(0)
        if Revise(csp, xi, xj):
            if len(csp.values[xi]) == 0:
                return False
            for peer in csp.peers[xi]:
```

```
                if peer != xj:
                    queue.append((peer, xi))
    return True

def Revise(csp, xi, xj):
    revised = False
    for value in csp.values[xi]:
        if not any(v != value for v in csp.values[xj]):
            csp.values[xi] = csp.values[xi].replace(value, '')
            revised = True
    return revised
```

Key insight

The Revise function removes a value v from domain(xi) only when domain(xj) = {v} (i.e., xj is already assigned to v). This ensures no two peers can hold the same value.

Complexity

- Time: $O(n^2 \cdot d^3)$ where n = number of variables (81) and d = max domain size (9)
- Space: $O(n^2)$ for the queue

3.2 Backtracking Search

Backtracking Search is a depth-first search that assigns values to one variable at a time and backtracks when an inconsistency is detected.

Implemetation

```
def Backtracking_Search(csp):
    assignment = {}
    for var in csp.variables:
        if len(csp.values[var]) == 1:
            assignment[var] = csp.values[var]
    return Recursive_Backtracking(assignment, csp)

def Recursive_Backtracking(assignment, csp):
    if isComplete(assignment):
        return assignment

    var = Select_Unassigned_Variables(assignment, csp)

    for value in Order_Domain_Values(var, assignment, csp):
        if isConsistent(var, value, assignment, csp):
            assignment[var] = value
            # Save current domain values for restoration on backtrack
```

```
        saved_values = deepcopy(csp.values)
        inferences = Inference(assignment, {}, csp, var, value)
        if inferences != "FAILURE":
            result = Recursive_Backtracking(assignment, csp)
            if result != "FAILURE":
                return result
        # Undo: remove assignment and restore domains
        del assignment[var]
        csp.values = saved_values
return "FAILURE"
```

3.3 Heuristics and Optimizations

3.3.1 Minimum Remaining Values (MRV) Heuristic

The MRV heuristic (also called "most constrained variable" or "fail-first") selects the unassigned variable with the smallest domain:

```
def Select_Unassigned_Variables(assignment, csp):
    unassigned_variables = dict(
        (sq, len(csp.values[sq])) for sq in csp.values
        if sq not in assignment.keys()
    )
    mrv = min(unassigned_variables, key=unassigned_variables.get)
    return mrv
```

Rationale: By choosing the variable most likely to cause a failure, the algorithm detects dead-ends earlier, significantly pruning the search tree.

3.3.2 Forward Checking

After each assignment, Forward Checking propagates the constraint to all peers by removing the assigned value from their domains:

```
def Inference(assignment, inferences, csp, var, value):
    inferences[var] = value
    for neighbor in csp.peers[var]:
        if neighbor not in assignment and value in
csp.values[neighbor]:
            if len(csp.values[neighbor]) == 1:
                return "FAILURE"  # would create empty domain
            remaining = csp.values[neighbor] =
csp.values[neighbor].replace(value, "")
            if len(remaining) == 1:
                flag = Inference(assignment, inferences, csp,
neighbor, remaining)
```

```
                if flag == "FAILURE":
                    return "FAILURE"
    return inferences
```

Key features:

- Detects failures early when a peer's domain would become empty.
- Performs recursive propagation: when a domain is reduced to a single value, that value is further propagated to its peers (creating a chain reaction effect similar to constraint propagation).

3.3.3 Domain Restoration on Backtrack

When backtracking, all domain changes made during inference are reversed by restoring a deep copy of the domain values:

```
saved_values = deepcopy(csp.values)
# ... attempt assignment and inference ...
# On failure:
del assignment[var]
csp.values = saved_values  # restore all domains
```

4. GUI Features

The interactive Pygame-based GUI provides:

- Board Visualization: 9×9 grid with color-coded cells (clues = white, AC-3 assignments = green, guesses = blue, conflicts = red).
- Pencil Marks: Remaining domain values displayed as small digits in a 3×3 sub-grid within each cell.
- Algorithm Selection: Choose between "AC-3 + Backtracking", "Backtracking Only", or "AC-3 Only".
- Speed Control: Adjustable step delay slider (5–500ms) for controlling animation speed.
- Step-by-Step Mode: Pause/Resume and single-step controls for detailed algorithm inspection.
- Batch Mode: "Solve All" button to solve all puzzles in a file and export results.
- Metrics Panel: Real-time display of time elapsed, nodes explored, guesses made, and backtracks.

5. Experimental Results

5.1 Test Datasets

Dataset	Puzzles	Avg Clues	Min Clues	Max Clues
Project Euler	50	28.4	22	36
Magic Tour	95	20.6	17	26

The Magic Tour dataset contains significantly harder puzzles with fewer given clues (average 20.6 vs 28.4), making them more challenging for constraint-based solvers.

5.2 Benchmark Results

The solver was benchmarked using the combined AC-3 + Backtracking algorithm. All tests were run on the same machine.

5.2.1 Overall Performance Summary

Metric	Euler	Magic Tour
Total Puzzles	50	95
Solved	50/50 (100%)	95/95 (100%)
Total Time	2.09s	86.14s
Average Time	0.042s	0.907s
Median Time	0.041s	0.160s
Min Time	0.022s	0.040s
Max Time	0.070s	13.846s

The solver achieves a 100% solve rate across all 145 puzzles in both datasets, demonstrating the completeness of the Backtracking Search algorithm.

5.2.2 Time Distribution

Time Range	Euler	Magic Tour
< 50ms	42 (84%)	5 (5.3%)
50 – 500ms	8 (16%)	69 (72.6%)
500ms – 2s	0 (0%)	10 (10.5%)
> 2s	0 (0%)	11 (11.6%)

6. Analysis and Discussion

6.1 Effect of Number of Clues on Difficulty

The experimental results clearly demonstrate an inverse correlation between the number of given clues and solving time:

Clue Count	Euler Average Time	Magic Tour Average Time
17	-	2.99s
19 – 22	0.063s	0.18s
23 – 26	0.048s	0.13s
27 – 30	0.044s	-
31 – 36	0.036s	-

Observations:

- Puzzles with 17 clues (Magic Tour) average ~64× longer solving time compared to puzzles with 30+ clues (Euler).
- The Euler dataset, with an average of 28.4 clues, contains relatively moderate puzzles where AC-3 alone can significantly reduce domains.
- The Magic Tour dataset pushes the solver harder: 46 out of 95 puzzles have only 17 clues, forcing extensive backtracking.

6.2 Role of AC-3 Preprocessing

AC-3 preprocessing dramatically reduces the search space before backtracking begins:

1. For easy puzzles (many clues): AC-3 alone can often solve the entire puzzle by repeatedly propagating singleton domains, eliminating the need for backtracking entirely.
2. For hard puzzles (few clues): AC-3 significantly reduces variable domains (e.g., from 9 candidates to 2–4), which:
    - Reduces the branching factor of the search tree
    - Enables MRV to make more informed variable selections
    - Decreases the number of guesses and backtracks needed

6.3 Effectiveness of MRV Heuristic

The MRV (Minimum Remaining Values) heuristic is critical for performance. By always selecting the variable with the fewest remaining domain values:

- Fail-first principle: Variables most likely to cause a conflict are tried first, detecting dead-ends early.
- Reduced branching: Variables with smaller domains have fewer values to try, decreasing the effective branching factor.

- Without MRV (e.g., using a fixed ordering), the hardest Magic Tour puzzles could take orders of magnitude longer.

6.4 Forward Checking with Recursive Propagation

The Inference function goes beyond basic forward checking by implementing recursive constraint propagation:

1. When a value is assigned to a variable, it is removed from all peers' domains.
2. If a peer's domain is reduced to a single value, that value is immediately propagated to its peers.
3. This creates a cascade effect similar to running AC-3 locally after each assignment.

This is particularly effective for Sudoku because assigning one cell often forces a chain of singleton assignments across the board.

6.5 Performance Comparison: Euler vs. Magic Tour

Aspect	Euler	Magic Tour
Average Clues	28.4	20.6
Average Solve Time	41.7ms	906.7ms
Time Variance	Very low ($\sigma \approx$ 10ms)	Very high ($\sigma \approx$ 2.3 s)
AC-3 alone sufficient?	Often	Rarely
Backtracking depth	Shallow	Often deep
Worst-case Time	70ms	13.8s

The Magic Tour dataset's high variance is notable: the median time is only 160 ms, but the worst case is 13.8 s — an 87× ratio, compared to only 3.3× for Euler. This reflects the exponential nature of backtracking search on truly hard instances.

7. Conclusion

7.1 Summary of Results

- The Sudoku Solving Agent successfully solves all 145 test puzzles (50 Euler + 95 Magic Tour) with a 100% solve rate.
- The combined AC-3 + Backtracking approach with MRV heuristic and Forward Checking provides an efficient and complete solver.
- Project Euler puzzles (avg 28.4 clues) are solved in an average of 42 ms.
- Magic Tour puzzles (avg 20.6 clues) are solved in an average of 907 ms, with the hardest puzzle (17 clues) taking 13.8 s.

7.2 Key Takeaways

1. Constraint propagation (AC-3) is essential — it dramatically reduces the search space before backtracking begins.
2. MRV heuristic is critical — choosing the most constrained variable first prevents exponential blowup on hard puzzles.
3. Forward Checking with recursive propagation acts as an embedded mini-AC-3 during search, catching inconsistencies early.
4. Puzzle difficulty scales inversely with clue count — fewer clues mean larger search spaces and more backtracking.
5. The algorithm is complete — guaranteed to find a solution if one exists, or correctly report failure.

7.3 Potential Improvements

Improvement	Expected Benefit
Least Constraining Value (LCV) ordering	Try values that rule out of fewest options for peers first
Naked/Hidden Pairs/Triples	Additional constraint propagation techniques
Constraint Learning	Remember and avoid previously discovered dead-end patterns
MAC (Maintaining Arc Consistency)	Run full AC-3 after each assignment instead of just forward checking
Bitwise domain representation	Use integer bitmasks instead of strings for faster domain operations

8. Personal Reflection

This assignment was both challenging and rewarding. I particularly enjoyed modeling Sudoku as a formal CSP and seeing how classical AI techniques — AC-3 and Backtracking Search — come together to solve puzzles that would be tedious to tackle by hand. Building the interactive GUI with real-time visualization was one of the most engaging parts, as it allowed me to observe the algorithms in action: watching the solver prune domains, make guesses, and backtrack in real time gave me a much deeper intuition for how constraint propagation and search interact. The most difficult aspect was implementing the Forward Checking with recursive propagation correctly — ensuring that domain changes are properly undone on backtrack required careful handling of deep copies and state restoration. Debugging subtle off-by-one errors in domain manipulation was also time-consuming. Overall, this assignment strengthened my understanding of CSPs, heuristic search, and the importance of preprocessing (AC-3) in reducing computational effort.

9. Time Estimation

Phase	Time
Understanding the CSP formulation and algorithms	0.5 hour
Implementing core solver (AC-3, Backtracking, MRV, Forward Checking)	1.5 hours
Building the interactive GUI with Pygame	1 hour
Testing, debugging, and benchmarking	1 hour
Writing the report and analysis	1 hour
Total	5 hours

Appendix A: Output Files

- EulerSolutions.txt — 50 solved puzzles, each as an 81-character string
- MagicTourSolutions.txt — 95 solved puzzles, each as an 81-character string
- benchmark.py — Script used to collect the experimental statistics presented in this report

Appendix B: How to Run

```
# Command-line batch solver
python sudoku.py --inputFile data/euler.txt
# Interactive GUI
python gui.py
# Benchmark script
python benchmark.py
```